

Chase A. Bonfiglio

(757) 663-6188 | bonfigliochase@gmail.com | chasebonfiglio.com | linkedin.com/in/chasebonfiglio

EDUCATION

Virginia Polytechnic Institute and State University

B.S. Mechanical Engineering, Robotics & Mechatronics, GPA: 3.57

Available: Summer 2027 internship | Spring or Fall 2027 co-op

Expected 2028

Blacksburg, VA

RESEARCH

Intelligent Materials for Biomimetic Sensors & Actuators Lab

Undergraduate Researcher, Dr. Scott Kennedy, Hume Center

2025 – Present

Blacksburg, VA

- Investigating whether K-group recruitment scheduling, firing fresh shape memory alloy (SMA) wires while spent ones cool, breaks the single-wire thermal bandwidth ceiling in a biomimetic ankle joint
- Designed and built a 12-wire Flexinol SMA bank in antagonistic configuration, with individually switched MOSFET drive channels and continuous zigzag routing for self-equalizing tension
- Instrumented the rig for per-channel resistance measurement with an NI USB-6451 DAQ and 74HC4053 analog multiplexers, enabling self-sensing of thermal recovery without added sensors, and wrote the embedded firmware driving it: gated firing, maximum on-time enforcement, watchdog protection, and automated test sequencing
- Presented at the Virginia Tech Hume Colloquium; first-author paper in preparation for IEEE RoboSoft 2027

ARC-AGI-3 Reasoning Harness

2026

Independent Research

- Hand-built an agentic harness putting a small open-weight LLM in control of novel interactive environments, following Chollet's *On the Measure of Intelligence*: the harness renders, diffs, and executes, the model reasons. Restores core-knowledge priors the model lacks (objectness, number, geometry) without injecting task knowledge. Writeup at chasebonfiglio.com/arcagithree.html
- Training a custom Qwen3.8-27B model for mid chain-of-thought tool calling and recursive self-calls: the model pauses generation, fetches environment state through a read-only grammar, and resumes. Built against a measured failure where roughly 85% of chain tokens go to re-deriving known facts; models rarely open the channel unprompted, but one seeded call is self-sustaining
- Probed the model for Gestalt perceptual priors: figure-ground and similarity present, closure, object-unity, and continuity absent. Oracle-injection ablation moved goal inference from 0/5 to 4/5, confirming missing structure-noticing as the bottleneck. Finetuning against those gaps with leave-family-out evaluation, restricted to task-agnostic priors

ENGINEERING EXPERIENCE

MachWorks Engineering Design Team

UAV Avionics / Camera Tracking

2025 – 2026

Blacksburg, VA

- Developed a Ronin-S based autonomous gimbal system for UAV target tracking, with a ROS 2 framework that processed telemetry and issued real-time control over SBUS, and designed the vision-to-gimbal control architecture around Kalman filter state estimation to compensate for latency, dropped telemetry, and sensor tolerances
- Wrote the servo harness test suite and ran EMI testing; thermal and electrical characterization expected this fall

PROJECTS

SO-101 Bimanual Manipulation Platform | 3D Printing, Servo Control, Imitation Learning

2026

- Building the open-source SO-101 leader-follower teleoperation rig, 12 actuators across two arms, from self-printed structural parts
- Designing a compliant TPU gripper and training a policy to stack spools of wire

Underwater ROV and Telemetry System | SolidWorks, RS-485, Embedded C

2024 – 2025

- Designed and built a remotely operated underwater vehicle with PVC structure and bilge thrusters, controlled over an RS-485 telemetry link via PS2 controller across a 30 ft tether

TECHNICAL SKILLS

Languages & Tools: Python, PyTorch, Embedded C / Arduino, C++, MATLAB, ROS 2, Git, SolidWorks

Technical: Control algorithms, state estimation, sensor fusion, DAQ instrumentation, embedded systems, hardware-in-the-loop testing, model finetuning

Certifications: Certified SolidWorks Associate (CSWA); Certified SolidWorks Professional in Mechanical Design (CSWP)

LEADERSHIP

Pool Operator, ABRC

Lifeguard, Manager, Head Manager

2022 – 2026

Virginia Beach, VA

- Ran chemical and pump room operations, covering water chemistry balancing, filtration and circulation systems, chemical handling and storage, and health department inspections; promoted three times in four seasons while directing daily operations for a 30-person staff and training incoming assistant managers